

Lab 6

- (1) Give the statement of Rice's Theorem.
- (2) Prove Rice's Theorem (you can use the recursion theorem).
- (3) Use Rice's theorem to show that: C is a proper, nonempty subset of the class of all recursively enumerable languages. Then the following problem is undecidable: Given a Turing machine M , is $L(M) \in C$?
- (4) Give self-contained definitions of the following concept: an oracle TM, and Language A is Turing reducible to language B .
- (5) Show that $E_{TM} \leq_T A_{TM}$.